

Description

Students will learn more by using Construct 3 to make platformer games. They'll practice making simple levels and get really good at creating animation frames with sprite sheets.

Objectives

Students will:

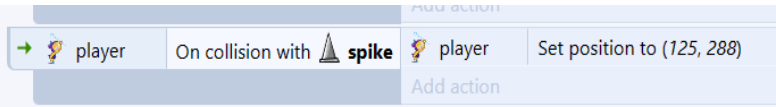
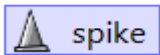
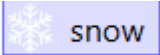
1. Recall the concept and elements of 2D Games.
2. Comprehend type of games : Platformer.
3. Comprehend the concept of sprite sheets and grasp the essence of animation frames.
4. Develop their own design.
5. Build a platformer game.

Tools

1. Construct 3

Class Structure

Duration	Topic
5 mins	Introduction
10 mins	Adding Spike
10 mins	Adding snowflake
10 mins	Behavior
10 mins	Control the player
15 mins	Collide the portal
10 mins	Get the candy and score
10 mins	Touch spike
5 mins	Update score
5 mins	Wrap - up

Planned Activities	Resources
Introduction(5 mins)	
Review : - Teacher asks students what they have learned from their previous module. - Teacher asks students if they still remember what they learned. - Tell the students, in this module they will learn how to create project platformer	Opening presentation link: here
Adding Spike (10 Mins)	
Show & Explain : Introduce to students that when the player touch with the spike, it will trigger the teleportation of the player back to starting position	 
Adding snowflake (10 Mins)	
Show & Explain : Let the students know we're planning to create an animation showing snow falling.	

- Add **particle object**
- Add the particle element and within its properties, select the image that should be displayed.
- Choose the snowflake and change:
 - Set it to 1, indicating the quantity of particles generated per second.
 - Spray cone: Adjust it to 20, representing the angle span within which particles are emitted.

Properties	
Rate	1
Spray cone	20
Type	Continuous spray ▼
Image	Edit
Object	snow ...
Initially visible	☒
Preview	☐

Behavior (10 Mins)

Show & Explain :

- Inform the students that they will be acquiring new behavior
- The behavior that students need to import :
 - Player : Bound to Layout, Platformer, Scroll to
 - Tiles : Solid

Player Behaviour :

Behaviors	
BoundToLayout	
Bound by	Edge ▼
Platform	
Max speed	330
Acceleration	1500
Deceleration	1500
Jump strength	500
Gravity	1500
Max fall speed	1000
Double-jump	☒
Jump sustain	0
Default controls	☒
Enabled	☒
ScrollTo	
Enabled	☒

Tiles Behavior :

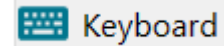
Behaviors	
Solid	
Enabled	☒
Tags	

Control the player (10 Mins)

Show & Explain :

- Inform the students to control the player we need Keyboard as Object
- Provide them with directions to include an event sheet so the player is able to move smoothly between the left and right sides.

Keyboard Object :



Keyboard Event Sheet :

Keyboard	← is down	player	Set Mirrored
		Add action	
Keyboard	→ is down	player	Set Not mirrored
		Add action	

Collide the portal (15 Mins)

Show & Explain :

- Explain to students that when the player comes into contact with the portal, it will trigger a relocation of the player to a different position.
- And the player will redirect to the finish line.

player	On collision with portal	System	Wait 0.5 seconds
		player	Set position to Sprite (image point 0)
		player	Set animation to "Animation 2" (play from beginning)
		Add action	

Get the candy and score (10 Mins)

Show & Explain :

- Inform the students that when the player comes into contact with the candy, the candy will vanish on its own.
- Once the candy disappears, the score will be incremented.
- Give the students instruction to create global number candy (**Variable**)

System	Every tick	Text	Set text to "Candy: "&candy
		Text	Set position to 🍬 player (image point 1)
Add action			

 *Global number* **candy = 0**

Touch spike (10 Mins)

Show & Explain :

- Provide guidance to the students that if the player makes contact with the spike, the player being sent back to the initial position.

→ 🍬 player	On collision with 📌 spike	🍬 player	Set position to (125, 288)
Add action			

Update score (10 Mins)

Show & Explain :

- Clarify that it's important to consistently update the score and display it using text.

Extra task :

- Add sound

Wrap Up (5 mins)

Wrapping Up

- We wrap up by asking questions to confirm students' grasp of the main

concepts. Raise your hands if you can explain what comparison operators and logical operators are.

- We express our gratitude to the students for their commendable effort throughout the session.

☐☐ Good Luck With The Class !! ☐☐